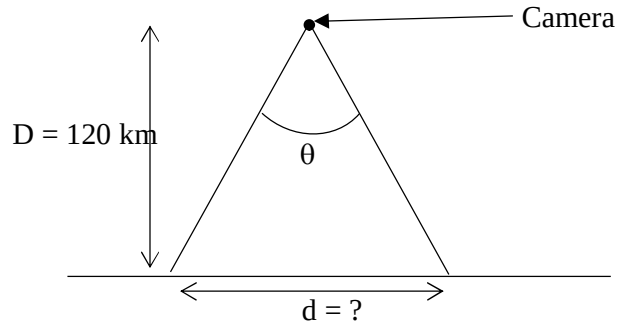


$\lambda = 1.3 \text{ m}$ is impossible.

17 B

Rayleigh's criterion



Just resolved: $\theta = \frac{\lambda}{a} \approx \frac{d}{D}$ when θ is very small.

$$\begin{aligned} d &= \frac{\lambda D}{a} \\ &= \frac{5.5 \times 10^{-7} \times 120 \times 10^3}{2.5 \times 10^{-2}} \\ &= 2.6 \text{ m} \end{aligned}$$

18 A